SQL Aggregate Functions & Operations
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Course: Data Science

Question 1

```
SELECT DISTINCT department
FROM students;
```

department
IT
HR
Finance

Question 2.

```
SELECT department, AVG (age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

Question 3.

```
SELECT department, COUNT () AS student_count
FROM students
GROUP BY department
~~Having~~ HAVING COUNT () >1;
```

department	Student_count
IT	2
HR	2

Question 4

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Question 5

```
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

Question 7

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Question 9

```
SELECT MAX(grade) AS max_grade,
MIN(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84.6

Question 10

```
SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Question 11

```
SELECT department, SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	177000	10500
HR	109000	7500
Finance	70000	6000

Question 12

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

Question 13

```
SELECT employee_id, name, salary, bonus, (salary + bonus)
AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Question 14

```
SELECT department,
SUM(budget) AS total_budget, AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

Question 15

```
SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120000
3	Payroll System	Finance	80000
5	HR Portal	HR	50000